

```
In [1]: # data manipulation
import pandas as pd
import numpy as np

#import attricion data
df=pd.read_csv('C:\\Users\\Public\\Desktop\\DAT-430\\HR Attrition Data.csv')
```

```
In [2]: #inspect the data
df.head()
```

Out[2]:

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education
0	38	Yes	Travel_Rarely	1180	Research & Development	29	1
1	49	No	Travel_Frequently	279	Research & Development	8	1
2	28	Yes	Non-Travel	1366	Research & Development	24	2
3	33	No	Travel_Frequently	1392	Research & Development	3	4
4	27	No	Travel_Rarely	591	Research & Development	2	1

5 rows × 35 columns



In []:

```
In [17]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
 #   Column                                  Non-Null Count  Dtype
---  -
 0   Age                                    1470 non-null   int64
 1   Attrition                             1470 non-null   object
 2   BusinessTravel                         1470 non-null   object
 3   DailyRate                             1470 non-null   int64
 4   Department                             1470 non-null   object
 5   DistanceFromHome                      1470 non-null   int64
 6   Education                             1470 non-null   int64
 7   EducationField                         1470 non-null   object
 8   EmployeeCount                         1470 non-null   int64
 9   EmployeeNumber                        1470 non-null   int64
10   EnvironmentSatisfaction                1470 non-null   int64
11   Gender                                 1470 non-null   object
12   HourlyRate                            1470 non-null   int64
13   JobInvolvement                        1470 non-null   int64
14   JobLevel                              1470 non-null   int64
15   JobRole                                1470 non-null   object
16   JobSatisfaction                       1470 non-null   int64
17   MaritalStatus                         1470 non-null   object
18   MonthlyIncome                         1470 non-null   int64
19   MonthlyRate                           1470 non-null   int64
20   NumCompaniesWorked                    1470 non-null   int64
21   Over18                                1470 non-null   object
22   OverTime                              1470 non-null   object
23   PercentSalaryHike                     1470 non-null   int64
24   PerformanceRating                     1470 non-null   int64
25   RelationshipSatisfaction                1470 non-null   int64
26   StandardHours                         1470 non-null   int64
27   StockOptionLevel                      1470 non-null   int64
28   TotalWorkingYears                     1470 non-null   int64
29   TrainingTimesLastYear                 1470 non-null   int64
30   WorkLifeBalance                       1470 non-null   int64
31   YearsAtCompany                        1470 non-null   int64
32   YearsInCurrentRole                    1470 non-null   int64
33   YearsSinceLastPromotion                1470 non-null   int64
34   YearsWithCurrManager                   1470 non-null   int64
dtypes: int64(26), object(9)
memory usage: 402.1+ KB
```

```
In [4]: #remove columns
df.drop(['DailyRate',
        'EmployeeCount',
        'EmployeeNumber',
        'Over18'], axis=1, inplace=True)
```

```
In [7]: df.describe()
```

```
Out[7]:
```

	Age	DistanceFromHome	Education	EnvironmentSatisfaction	HourlyRate
count	1470.000000	1470.000000	1470.000000	1470.000000	1470.000000
mean	36.923810	9.192517	2.912925	2.721769	65.891156
std	9.135373	8.106864	1.024165	1.093082	20.329428
min	18.000000	1.000000	1.000000	1.000000	30.000000
25%	30.000000	2.000000	2.000000	2.000000	48.000000
50%	36.000000	7.000000	3.000000	3.000000	66.000000
75%	43.000000	14.000000	4.000000	4.000000	83.750000
max	60.000000	29.000000	5.000000	4.000000	100.000000

8 rows × 23 columns

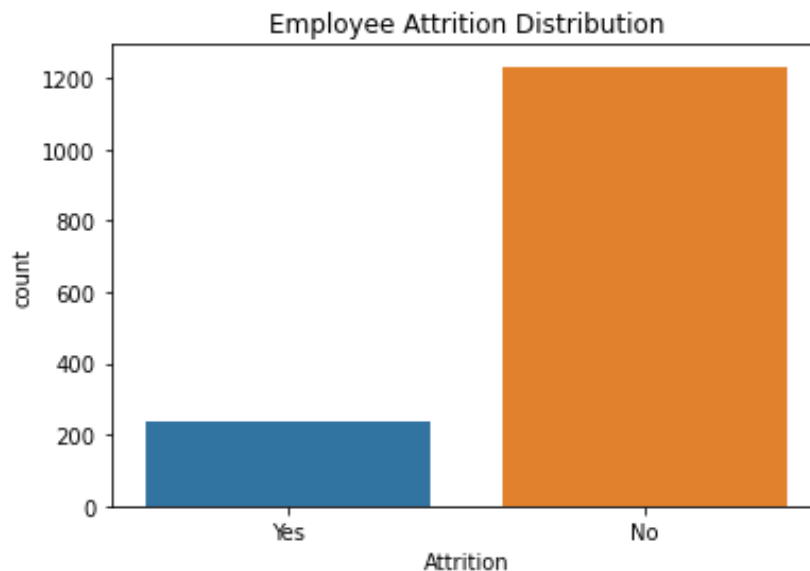


```
In [6]: #understanding attrition distribution
df["Attrition"].value_counts(normalize=True)
```

```
Out[6]: No      0.838776
Yes       0.161224
Name: Attrition, dtype: float64
```

```
In [21]: import seaborn as sns
import matplotlib.pyplot as plt

sns.countplot(data=df, x='Attrition')
plt.title('Employee Attrition Distribution')
plt.show()
```



```
In [8]: #changing Attrition from object to numbers
df['Attrition']=df['Attrition'].map({"No":0,"Yes":1})
```

```
In [9]: import matplotlib.pyplot as plt

corr=df.corr()
plt.figure(figsize=(12,8))
sns.heatmap(corr[['Attrition']].sort_values(by='Attrition', ascending=False), annot=True)
plt.title('Correlation with Attrition')
plt.show
```

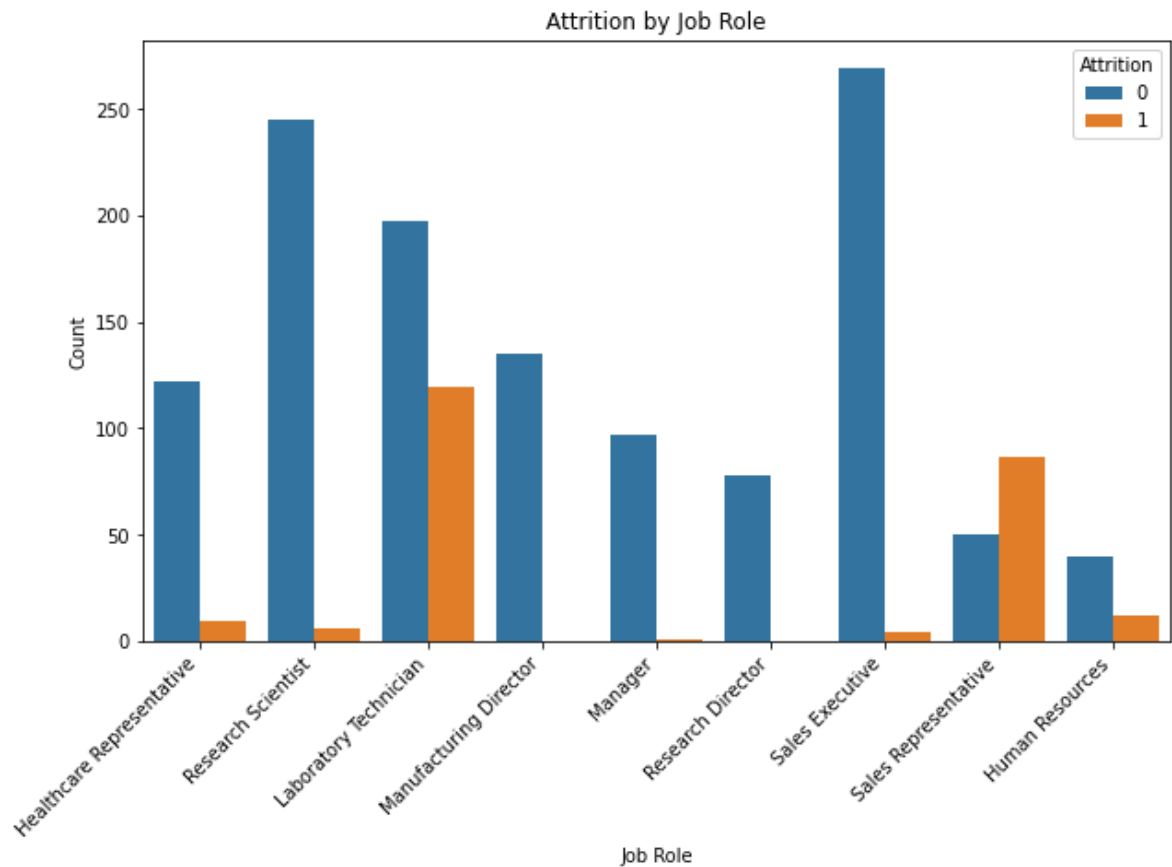
```
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---
NameError                                Traceback (most recent call last)
<ipython-input-9-b5faf960435f> in <module>
      3 corr=df.corr()
      4 plt.figure(figsize=(12,8))
----> 5 sns.heatmap(corr[['Attrition']].sort_values(by='Attrition', ascending=False), annot=True)
      6 plt.title('Correlation with Attrition')
      7 plt.show

NameError: name 'sns' is not defined

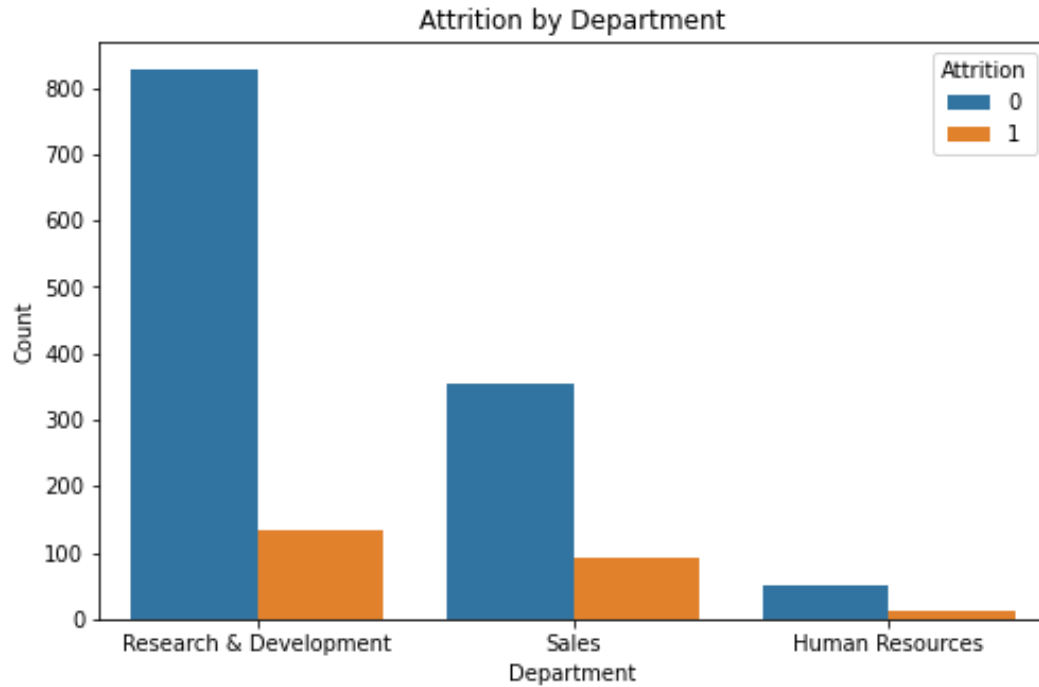
<Figure size 864x576 with 0 Axes>
```

```
In [10]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Visualizing attrition by Job Role
plt.figure(figsize=(10,6))
sns.countplot(data=df, x='JobRole', hue='Attrition')
plt.title('Attrition by Job Role')
plt.xticks(rotation=45, ha='right')
plt.xlabel('Job Role')
plt.ylabel('Count')
plt.show()
```



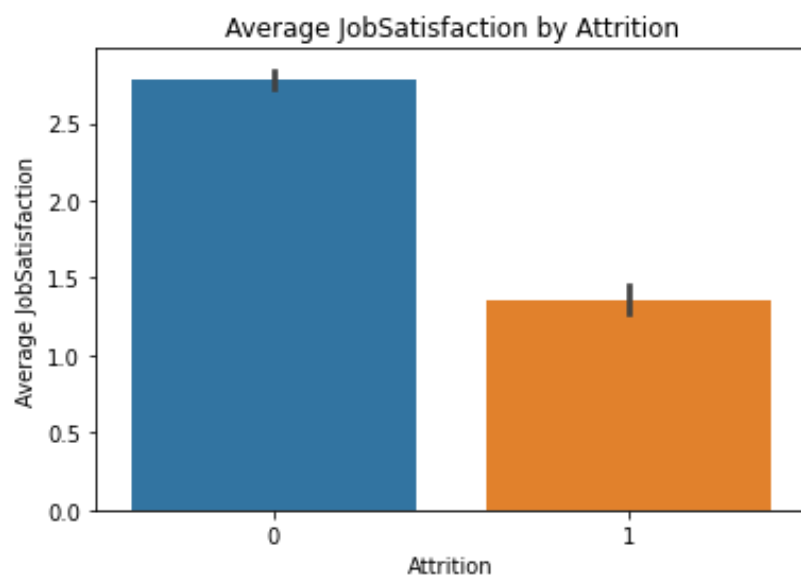
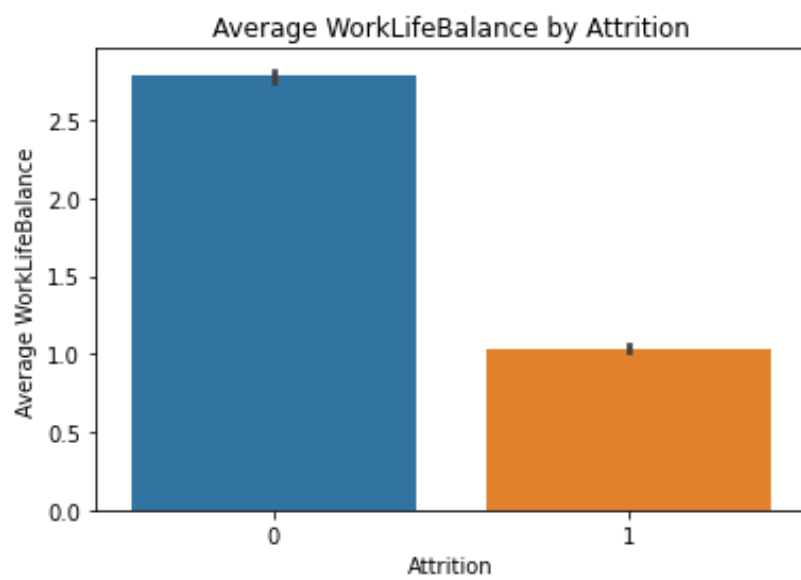
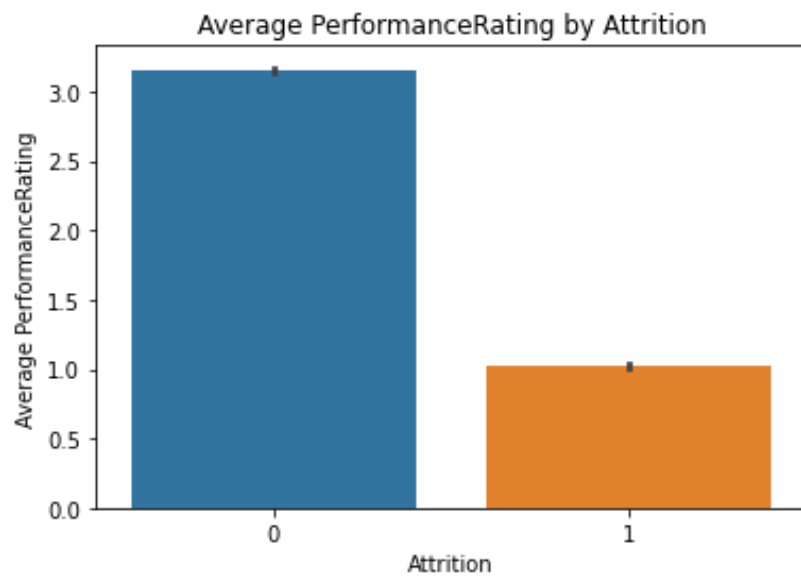
```
In [11]: # Visualizing attrition by Department
plt.figure(figsize=(8,5))
sns.countplot(data=df, x='Department', hue='Attrition')
plt.title('Attrition by Department')
plt.xlabel('Department')
plt.ylabel('Count')
plt.show()
```



In [12]: *#visualizing top three negative Attrition*

```
cols = ['PerformanceRating', 'WorkLifeBalance', 'JobSatisfaction']

for col in cols:
    plt.figure(figsize=(6, 4))
    sns.barplot(data=df, x='Attrition', y=col)
    plt.title(f'Average {col} by Attrition')
    plt.ylabel(f'Average {col}')
    plt.xlabel('Attrition')
    plt.show()
```



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